Algebraic Topology 3

Labix

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Abstract

References:

- Notes on Algebraic Topology by Oscar Randal-Williams:
  The first chapter gives a complete treatment of the first three sections of these notes, as well as providing the importance of fibrations on the higher homotopy groups. These notes are highly recommended to understanding the first three sections.
- Algebraic Topology by Allen Hatcher:
  A more or less complete dictionary on all topics of these notes. However it is prone to the same problem in the sense that Hatcher's book is rather terse and definitions and parts of some theorems are scattered throughout the paragraphs rather than having a complete statement for reference. Nevertheless it is still the standard reference of the notes, albeit organized in a slightly different way.
- A non-visual proof that higher homotopy groups are abelian by Shintaro Fushida-Hardy:
  This short piece of article proves that the higher homotopy groups are abelian in a purely algebraic way. Most geometric visualization of such a proof has the same underlying idea as the algebraic method.

Contents

1 The Homotopy Excision Theorem

1.1 Proof of the Main Theorem

Theorem 1.1.1 (The Homotopy Excision Theorem)
Let X be a CW-complex and A, B be sub complexes such that $X = A \cup B$ and $A \cap B \neq \emptyset$. If $(A, A \cap B)$ is m-connected and $(B, A \cap B)$ is n-connected for $m, n \geq 0$, then the map

$$\pi_k(\iota) : \pi_k(A, A \cap B) \to \pi_k(X, B)$$

induced by the inclusion $\iota : (A, A \cap B) \to (X, B)$ is an isomorphism for $0 \leq k < m + n$ and a surjection for $k = m + n$.

Proposition 1.1.2 Let (X, A) be a pair of r-connected CW complexes and let A be s-connected. Then the map

$$p_* : \pi_k(X, A) \to \pi_k(X/A)$$

induced by the quotient map $p : X \to X/A$ is an isomorphism for $0 \leq k \leq r + s$ and a surjection for $k = r + s + 1$.

1.2 Freudenthal Suspension Theorem

Theorem 1.2.1 (Freudenthal Suspension Theorem)
Let X be an n-connected CW complex. Then the following are true regarding the suspension map

$$\Sigma : \pi_k(X) \to \pi_{k+1}(\Sigma X)$$

- It is an isomorphism for $0 \leq k \leq 2n$.
- It is a surjection for $k = 2n + 1$.

Corollary 1.2.2
Let X be an n-connected CW complex. Then the following are true.
- $\pi_k(\Sigma X) \cong \pi_{k-1}(X)$ for $1 \leq k \leq 2n + 1$.
- ΣX is $(n + 1)$-connected.

Corollary 1.2.3
Let X be an n-connected CW complex. Let $k \in \mathbb{N}$. Then the sequence

$$\pi_k(X) \to \pi_{k+1}(\Sigma X) \to \cdots \to \pi_{k+r}(\Sigma^r X) \to \cdots$$

of abelian groups satisfy the following.
- If $k \leq 2n$, then the sequence is constant.
- If $k > 2n$, then the sequence is constant after $r = k - 2n$.

Proof
Suppose that $k \leq 2n$. By Freudenthal suspension theorem, if $k \leq 2n$ then $\pi_k(X) \cong \pi_{k+1}(\Sigma X)$. Then ΣX is $(n + 1)$-connected hence $\pi_{k+1}(\Sigma X) \cong \pi_{k+2}(\Sigma^2 X)$ is an isomorphism since $k + 1 \leq 2n + 2$. More generally, for $r \in \mathbb{N}$, $\Sigma^r X$ is $(r + n)$-connected hence

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism since $k + r \leq 2n + 2r$.

Now if $k > 2n$, then there exists $r \in \mathbb{N}$ such that $k + r \leq 2n + 2r$. Such an r is given by say $k - 2n$. Then by Freudenthal suspension theorem,

$$\pi_{k+r}(\Sigma^r X) \cong \pi_{k+r+1}(\Sigma^{r+1} X)$$

is an isomorphism. More generally, for $m \in \mathbb{N}$, $\Sigma^{r+m}X$ is $(r+m+n)$-connected hence

$$\pi_{k+r+m}(\Sigma^{r+m}X) \cong \pi_{k+r+m+1}(\Sigma^{r+m+1}X)$$

is an isomorphism since $k+r+m \leq 2n+2r+2m$. ■

Corollary 1.2.4
We have

$$\pi_n(S^n) \cong \mathbb{Z}$$

for all $n \in \mathbb{N}$.

Proof
Choose $X = S^1$ in Freudenthal suspension theorem. It is 0-connected and so we have a surjection $\pi_1(S^1) \to \pi_2(S^2)$ is a surjection by Freudenthal suspension theorem. Since $\pi_1(S^1) \cong \mathbb{Z}$, we conclude that either $\pi_2(S^2) \cong \mathbb{Z}$ or $\pi_2(S^2) \cong \mathbb{Z}/d\mathbb{Z}$ for some $d \geq 1$. But since $H_2(S^2) \cong \mathbb{Z}$, this implies that a map $S^2 \to S^2$ of degree i for all $i \in \mathbb{Z}$ exists. But maps of different degrees are not homotopic. Hence $\pi_2(S^2) \cong \mathbb{Z}$.

Now choose $X = S^2$ in Freudenthal suspension theorem. It is 1-connected. By corollary 1.2.3, we deduce that the sequence

$$\pi_2(S^2) \to \pi_3(S^3) \to \cdots \to \pi_n(S^n) \to \cdots$$

is constant. Hence $\pi_n(S^n) \cong \mathbb{Z}$. ■

Example 1.2.5
Let $n \in \mathbb{N}$. Then we have

$$\pi_n\left(\bigvee_{i\in I} S^n\right) \cong \bigoplus_{i\in I} \pi_n(S^n) \cong \mathbb{Z}^{\oplus|I|}$$

given by factoring a map $S^n \to \bigvee_{i\in I} S^n$ into its components.

Proof
Suppose first that I is finite. Notice that $\prod_{i\in I} S^n$ has cells in dimensions a multiple of n. Hence $(\prod_{i\in I} S^n \mid \bigvee_{i\in I} S^n)$ is $(2n-1)$-connected. By the long exact sequence of relative homotopy groups, we deduce an isomorphism

$$\pi_n\left(\bigvee_{i\in I} S^n\right) \cong \pi_n\left(\prod_{i\in I} S^n\right) \cong \bigoplus_{i\in I} \pi_n(S^n)$$

Now if I is not finite, consider the map $\bigoplus_{i\in I} \pi_n(S^n) \to \pi_n\left(\bigvee_{i\in I} S^n\right)$ given by composition $S^n \to S^n \hookrightarrow \bigvee_{i\in I} S^n$. This map is surjective since any map $S^n \to \bigvee_{i\in I} S^n$ has compact image because S^n is compact so that this map factors through a finite wedge, and we have proven such a map comes from $\bigoplus_{i\in I} \pi_n(S^n)$ in the finite case. Similarly, f lies in the image of the map and is null homotopic, the null homotopy must lie in a finite wedge, and we have proven in the finite case that the f must be null homotopic along each factor in $\bigoplus_{i\in I} \pi_n(S^n)$. ■

Example 1.2.6
Let $n \geq 2$. Then we have

$$\pi_n(S^1 \vee S^n) \cong \mathbb{Z}[t, t^{-1}]$$

Example 1.2.7
Let X be the CW complex given by attaching an $(n+1)$-cell by the attaching map $\varphi : S^n \to$

$\bigvee_{i \in I} S^n$. Then we have

$$\pi_k(X) \cong \begin{cases} \frac{\bigoplus_{i \in I} \pi_n(S^n)}{\langle [\varphi] \rangle} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

2 Properties of the Homotopy Groups

2.1 Freudenthal Suspension Theorem

Theorem 2.1.1 (Freudenthal Suspension Theorem)
Let X be an n-connected CW complex. Then for $0 \leq k \leq 2n$, the induced map

$$\Sigma_* : \pi_k(X) \to \pi_{k+1}(\Sigma X)$$

is an isomorphism. For $k = 2n+1$, Σ_* is a surjection.

We can keep on suspending the space and the maps. Indeed if X is n-connected then, by Freudenthal suspension theorem ΣX is $(n+1)$-connected. We can then apply the suspension theorem again on ΣX and we see that $\Sigma^2 X$ is $(n+2)$-connected.

The following theorem is also said to be the Freudenthal suspension theorem.

Theorem 2.1.2 Let Y be $(n-1)$-connected. Consider the reduced suspension functor $\Sigma : \mathbf{hTop}_* \to \mathbf{hTop}_*$. Then $\Sigma : [X, Y] \to [\Sigma X, \Sigma Y]$ is bijective if $\dim(X) < 2n-1$. Moreover, it is a surjection if $\dim(X) = 2n-1$.

Corollary 2.1.3 There is an isomorphism

$$\pi_{n+k}(S^n) \cong \pi_{n+k+1}(S^{n+1})$$

for all $n \geq k+2$.

2.2 Excision for Homotopy Groups

2.3 The Homotopy Groups of a Fiber Bundle

Theorem 2.3.1 Let $p : E \to B$ be a fiber bundle. Let $A \subseteq B$. Let $y_0 \in E$ and $p(y_0) = x_0$. Then there is an isomorphism

$$\pi_n(E, p^{-1}(A), y_0) \cong \pi_n(B, A, x_0)$$

given by the induced map p_* for all $n \geq 2$.

3 Loop Spaces and The Compact Open Topology

3.1 The Mapping Space

Definition 3.1.1 (Compact Open Functions)
Let X, Y be spaces. Let $K \subseteq X$ be compact. Let $U \subseteq Y$ be open. Define

$$V(K, U) = \{f : X \to Y \mid f \text{ is a map and } f(K) \subseteq U\}$$

Definition 3.1.2 (The Mapping Space)
Let X, Y be spaces. Define the mapping space of X and Y to be the set

$$\text{Map}(X, Y) = \{f : X \to Y \mid f \text{ is continuous }\}$$

together with the topology generated by the subbase

$$\{V(K, U) \mid K \subseteq X \text{ is compact and } U \subseteq Y \text{ is open }\}$$

Let X, Y, Z be sets. Recall that in Category Theory, there is an adjunction $- \times Y \dashv \text{Hom}_{\textbf{Set}}(Y, -)$. This means that there is a natural bijection of sets

$$\text{Hom}_{\textbf{Set}}(X \times Y, Z) \cong \text{Hom}_{\textbf{Set}}(X, \text{Hom}_{\textbf{Set}}(Y, Z))$$

When we equip the set of homomorphisms with the compact open topology, in general we lose the adjunction.

Proposition 3.1.3
Let X, Y, Z be spaces. Let $f : X \times Y \to Z$ be a function. If X is locally compact, then f is continuous if and only if the map $Y \to \text{Map}(X, Z)$ given by $y \mapsto f(-, y)$ is continuous.

One way to resolve this is to restrict the category of spaces to compactly generated and weakly Hausdorff spaces.

Definition 3.1.4 (The Pointed Mapping Space)
Let X, Y be pointed spaces. Define the mapping space of X and Y to be the set

$$\text{Map}_*(X, Y) = \{f : X \to Y \mid f \text{ is a pointed map }\}$$

together the topology given by ???

3.2 Loop Spaces

Definition 3.2.1 (Loop Spaces)
Let X be a pointed space. Define the loop space of X to be

$$\Omega X = \text{Map}_*(S^1, X)$$

Lemma 3.2.2
Let G be an abelian group and let $n \in \mathbb{N}$. Then there is a homeomorphism

$$\Omega K(G, n) \cong K(G, n - 1)$$

Definition 3.2.3 (Free Loopspace)
Let X be a space. Define the free loopspace of X to be

$$LX = \text{Map}(S^1, X)$$

4 H-Spaces

4.1 H-Spaces and the Hopf Algebra Structure on Cohomology

Definition 4.1.1 (H-Spaces)
Let X be a space. We say that X is a H-space if there exists a map $\mu : X \times X \to X$ and a point $e \in X$ such that the maps $x \mapsto \mu(x,e)$ and $x \mapsto \mu(e,x)$ are homotopic to the identity map.

Example 4.1.2
Let G be a topological group. Then G is a H-space.

Example 4.1.3
Let (X, x_0) be a pointed space. Then $\Omega(X, x_0)$ is a H-space.

Lemma 4.1.4
Let X be a H-space. Let Y be a space. Then $[Y, X]$ has multiplication and identity.

Proposition 4.1.5
Let X be a H-space with multiplication given by μ. Let R be a commutative ring. Suppose that the following are true.
- $H^0(X; R) \cong R$.
- $H^n(X; R)$ is a finitely generated free R-module for each $n \in \mathbb{N}$.

Then $H^*(X; R)$ is a Hopf algebra whose comultiplication is induced by μ^*.

4.2 Homotopy Associative H-Space and the Pontryagin Product

Definition 4.2.1 (Homotopy Associative H-Space)
Let X be a H-space. We say that X is homotopy associative if the two maps

$$(x,y,z) \mapsto \mu(x, \mu(y,z)) \quad \text{and} \quad (x,y,z) \mapsto \mu(\mu(x,y), z)$$

are homotopic.

Lemma 4.2.2
Let X be a space. Then X is a group like H-space if and only if X is a monoid object in $\mathbf{hTop}$.

Lemma 4.2.3
Let X be a homotopy associative H-space. Let Y be a space. Then $[Y, X]$ is a monoid.

Definition 4.2.4 (Pontryagin Product)
Let X be a homotopy associative H-space. Let R be a commutative ring. Define the Pontryagin product of X to be the composite

$$H_*(X; R) \otimes_R H_*(X; R) \xrightarrow{\times} H_*(X \times X; R) \xrightarrow{\mu_*} H_*(X; R)$$

where the first map is the cross product given by the Kunneth theorem.

Proposition 4.2.5
Let X be a homotopy associative H-space. Then the Pontryagin product gives a ring structure to the direct sum

$$H_*(X; R) = \bigoplus_{n=0}^{\infty} H_n(X; R)$$

4.3 Group Like H-Spaces

Definition 4.3.1 (Group Like H-Spaces)
Let X be a H-space. We say that X is group like if the following are true.
- X is homotopy associative.
- There exists a map $j : X \to X$ such that $\mu \circ (\text{id}, j)$ and $\mu \circ (j, \text{id})$ are homotopic to the identity map.

Lemma 4.3.2
Let X be a space. Then X is a group like H-space if and only if X is a group object in **hTop**.

Lemma 4.3.3
Let X be a group like H-space. Let Y be a space. Then $[Y, X]$ is a group.

Proposition 4.3.4
Let X be homotopy associative H-space. Then X is group like if and only if the map $(x, y) \mapsto (x, xy)$ is a homotopy equivalence.

Definition 4.3.5 (Homotopy Commutative H-Space)
Let X be a H-space. We say that X is homotopy commutative if the two maps

$$(x, y) \mapsto \mu(x, y) \quad \text{and} \quad (x, y) \mapsto \mu(y, x)$$

are homotopic.

Lemma 4.3.6
Let X be a space. Then X is a group like H-space if and only if X is an abelian group object in **hTop**.

Proposition 4.3.7
Let X be a pointed space. Then the following are true.
- ΩX is a group like H-space.
- $\Omega^2 X$ is group like homotopy commutative H-space.

4.4 Group Structure on the Set of Maps

Definition 4.4.1 (Group Structure on Loop Spaces) Let X be a space. Define a group structure on ΩX as follows. Let $\cdot : \Omega X \times \Omega X \to \Omega X$ be defined as the concatenation: $(f, g) \mapsto f \cdot g$.

Proposition 4.4.2 Let X, Y be spaces. Then the group structure on ΩY endows $[X, \Omega Y]_*$ with a group structure defined as follows. The binary operation $+ : [X, \Omega Y]_* \times [X, \Omega Y]_* \to [X, \Omega Y]_*$ is defined by

$$([f], [g]) \mapsto [f + g]$$

where $f + g : X \to \Omega Y$ is defined by $(f + g)(x) = f(x) \cdot g(x)$.

Proposition 4.4.3 Let X, Y be spaces. Then for $n \geq 2$, the group

$$[X, \Omega^n Y]_*$$

is abelian.

5 The Free Monoid of a Space

5.1 The James Construction

Definition 5.1.1 (James Reduced Product)
Let (X, e) be a pointed space.
- Define $J_n(X) = \frac{X^n}{\sim}$ where $(x_1, \dots, x_{i-1}, e, x_{i+1}, \dots, x_n) \sim (x_1, \dots, x_i, e, x_{i+2}, \dots, x_n)$.
- Define the James reduced product of (X, e) to be
$$J(X) = \text{colim}_{n\in\mathbb{N}} J_n(X)$$

Lemma 5.1.2
Let (X, e) be a CW complex where e is a 0-cell. Then $J(X)$ is a CW complex.

Proposition 5.1.3
Let X be a connected CW complex. Let R be a ring. Suppose that $H_*(X; R)$ is a free R-module. Then there is an isomorphism
$$H_*(J(X); R) \cong T(\widetilde{H}_*(X; R))$$

Let (X, x_0) be a pointed space. Recall that the suspension of X is defined to be
$$\Sigma X = \frac{X \times I}{X \times \partial I \cup \{x_0\} \times I}$$
Elements of ΣX look like (x, t) where $x \in X$ and $t \in I$. Since $(x, 0) \sim (x, 1)$ in the quotient, the map $t \mapsto (x, t)$ is a well defined loop in ΣX, hence an element of $\Omega\Sigma X$.

Proposition 5.1.4
Let (X, e) be a pointed connected CW complex where e is a 0-cell. Let $\lambda : J(X) \to \Omega\Sigma X$ be the map defined by $\lambda(x) = (t \mapsto (x, t))$. Then λ is a homotopy equivalence.

5.2 The Infinite Symmetric Product

Definition 5.2.1 (Infinite Symmetric Product)
Let (X, e) be a pointed space.
- Define $SP_n(X) = \frac{X^n}{S^n}$ the orbit space, where S^n acts on X^n by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.
- Define the infinite symmetric product of X to be
$$SP(X) = \text{colim}_{n\in\mathbb{N}} SP_n(X)$$
where the maps $SP_n(X) \hookrightarrow SP_{n+1}(X)$ is given by inclusion: $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_n, e)$.

Lemma 5.2.2
Let (X, e) be a pointed space. Then there is a homeomorphism
$$SP(X) \cong \text{colim}_{n\in\mathbb{N}} \frac{J_n(X)}{S_n}$$
where S_n acts on $J_n(X)$ by $\sigma \cdot (x_1, \dots, x_n) = (x_{\sigma(1)}, \dots, x_{\sigma(n)})$.

Theorem 5.2.3 (Dold-Thom Theorem)
Let (X, e) be a pointed space. Then there are isomorphisms
$$\widetilde{H}_n(X) \cong \pi_n(SP(X), e)$$
for each $n \in \mathbb{N}$.

Proposition 5.2.4
Let (X, e) be a pointed space. Then we have a factorization

$$\begin{tikzcd} \pi_n(X,e) \arrow[rr, "\text{incl}_*"] \arrow[dr, "h_n"'] & & \pi_n(\mathrm{SP}(X),e) \\ & \widetilde{H}_n(X) \arrow[ur, "\cong"'] & \end{tikzcd}$$

where h is the Hurewicz homomorphism.

6 Products between Homotopy Groups

6.1 The Samelson Product

Definition 6.1.1 (Samelson Product)
Let X, Y be spaces. Let G be a group like H-space. Let $f : X \to G$ and $g : Y \to G$ be maps. Define the Samelson product $[f, g] : X \wedge Y \to G$ by

$$[f, g] = [-, -] \circ (f \wedge g)$$

where $[-, -]$ is the commutator map on G.

Note that the Samelson product is well defined because the commutator map $[-, -] : G \times G \to G$ is trivial on $G \wedge G$ and so it descends to a map $G \wedge G \to G$.

Proposition 6.1.2
Let G be a group like H-space that is 0-connected. Then the following are true.
- The map $\pi_n(G) \times \pi_m(G) \to \pi_{n+m}(G)$ given by $([f], [g]) \mapsto [f, g]$ is a $\mathbb{Z}$-bilinear map.
- Let $[f] \in \pi_n(G)$ and $[g] \in \pi_m(G)$. Then we have

$$[f, g] = (-1)^{nm+1}[g, f]$$

Proposition 6.1.3 (The Super Jacobi Identity)
Let G be a group like H-space that is 0-connected. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{nk}[f, [g, h]] + (-1)^{nm}[g, [h, f]] + (-1)^{mk}[h, [f, g]] = 0$$

Corollary 6.1.4
Let G be a group like H-space that is 0-connected. Then $\pi_*(G)$ is a super Lie algebra.

6.2 The Whitehead Product

Definition 6.2.1 (The Whitehead Product)
Let X be a pointed space. Define the Whitehead product on X to be the map $[-, -]_{\text{Wh}} : \pi_n(X) \times \pi_m(X) \to \pi_{n+m-1}(X)$ as follows. For $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$, the Whitehead product is given by

$$[f, g]_{\text{Wh}} = (f \vee g) \circ \phi$$

where $\phi : S^{n+m-1} \to S^n \vee S^m$ is the attaching map of the $(n+m)$-cell in the CW complex structure of $S^n \times S^m$.

Proposition 6.2.2
Let X be a pointed space. Let τ be the isomorphism $\pi_{n+1}(X) \cong \pi_n(\Omega X)$. Let $[f] \in \pi_{n+1}(X)$ and $[g] \in \pi_{m+1}(X)$. Then we have

$$\tau([f, g]_{\text{Wh}}) = (-1)^{n-1}[\tau([f]), \tau([g])]$$

Proposition 6.2.3
Let X be a pointed space.
- The Whitehead product $[-, -]_{\text{Wh}}$ is a $\mathbb{Z}$-bilinear map.
- Let $[f] \in \pi_n(X)$ and $[g] \in \pi_m(X)$. Then we have

$$[f, g] = (-1)^{nm}[g, f]$$

Proposition 6.2.4 (The Super Jacobi Identity)

Let X be a pointed space. Let $[f] \in \pi_n(G)$, $[g] \in \pi_m(G)$ and $[h] \in \pi_k(G)$. Then we have

$$(-1)^{(n-1)(k-1)}[f,[g,h]] + (-1)^{(n-1)(m-1)}[g,[h,f]] + (-1)^{(m-1)(k-1)}[h,[f,g]] = 0$$

Corollary 6.2.5
Let X be a pointed space. Then $\pi_{*-1}(X)$ is a super Lie algebra.

6.3 The Hilton Milnor Theorem

Theorem 6.3.1 (Hilton-Milnor Theorem)
Let X, Y be connected spaces. Then there is a weak equivalence

$$\Omega\Sigma X \times \Omega\Sigma\left(\bigvee_{i=0}^{\infty}(Y \wedge X^{\wedge i})\right) \to \Omega\Sigma(X \vee Y)$$

Corollary 6.3.2
Let X, Y be connected spaces. Then there is a zigzag of weak equivalence

$$\Omega\Sigma(X \vee Y) \simeq \Omega\Sigma X \times \Omega\Sigma Y \times \Omega\Sigma\left(\bigvee_{i,j=1}^{\infty}(X^{\wedge i} \wedge Y^{\wedge j})\right)$$

given by twice applying the Hilton-Milnor theorem.

Corollary 6.3.3
There is a weak equivalence

$$\Omega\left(\Sigma X_1 \vee \cdots \vee \Sigma X_n\right) \to \prod_{w \in W} \Omega\Sigma\left(X_1^{\wedge w_{x_1}} \wedge \cdots \wedge X_n^{\wedge w_{x_n}}\right)$$

where W is a hall basis for the free Lie algebra on $x_1, \dots, x_n$ and w_{x_i} is the number of x_i appearing in w and the product is the product in **CGWH**.

7 Stable Phenomena

7.1 The Stable Homotopy Groups

In Algebraic Topology, we refer to a property or invariant being stable if applying the suspension functor to the space does not change the property (up to possibly a shift in index). The first such example is one we have already encountered.

Definition 7.1.1 (Stable Homotopy Groups) Let X be a space. Let $n \in \mathbb{N}$. Define the nth stable homotopy groups of X to be

$$\pi_n^s(X) = \operatorname*{colim}_{k\to\infty} \pi_{n+k}(\Sigma^k X)$$

This is well defined because of the Freudenthal suspension theorem, which states that the groups in the direct limit eventually stabilize. Indeed, the same theorem shows the following.

Proposition 7.1.2 Let X be a space. Then there is an isomorphism

$$\pi_n^s(X) \cong \pi_{n+1}^s(\Sigma X)$$

induced by the suspension functor between stable homotopy groups for any $n \in \mathbb{N}$.

Proof This can be seen by simply unwinding the definitions. We have that

$$\begin{aligned} \pi_{n+1}^s(\Sigma X) &= \operatorname*{colim}_{k\to\infty} \pi_{n+k+1}(\Sigma^{k+1} X) \\ &= \operatorname*{colim}_{k\to\infty} \pi_{n+k}(\Sigma^k X) \\ &= \pi_n^s(X) \end{aligned}$$

and so we conclude. ■

New: Graded ring structure on π_*^s.

Theorem 7.1.3 The stable homotopy groups define a collection of functors $\pi_n^s : \mathbf{CW}_* \to \mathbf{Ab}$ as follows.

- For X a pointed CW complex, $\pi_n^s(X)$ is the nth stable homotopy group of X
- For $f : X \to Y$ a map,
$$\pi_n^s(f) : \pi_n^s(X) \to \pi_n^s(Y)$$
is the image of f under the canonical map $[X, Y] \to [X, Y]^s$

Theorem 7.1.4 The stable homotopy functors $\pi_n^s : \mathbf{CW}_* \to \mathbf{Ab}$ for each $n \in \mathbb{N}$ defines a reduced homology theory.

This is untrue for the unstable homotopy groups. In particular, the fundamental group is not necessarily abelian.

7.2 Spanier Whitehead Duality

Recall that for $n \geq 2$, the set $[\Sigma^2 X, Z]$ for any spaces X and Z are abelian. Moreover, the suspension map $\Sigma : [\Sigma^2 X, Z] \to [\Sigma^3 X, \Sigma Z]$ is a group homomorphism. In particular, we can choose $Z = \Sigma^2 Y$. Now since the category $\mathbf{Ab}$ of abelian groups is cocomplete, the following inverse system

$$[\Sigma^2 X, \Sigma^2 Y] \xrightarrow{\Sigma} [\Sigma^3 X, \Sigma^3 Y] \xrightarrow{\Sigma} [\Sigma^4 X, \Sigma^4 Y] \longrightarrow \cdots$$

has an inverse limit. This leads to the following definition.

Definition 7.2.1 (Set of Stable Homotopy Classes of Maps) Let X and Y be space. The set of stable homotopy classes of maps from X to Y is defined to be the abelian group

$$[X,Y]^s = \operatorname*{colim}_{n\in\mathbb{N}\setminus\{0,1\}}[\Sigma^n X, \Sigma^n Y]$$

The following observation is crucial. When X and Y are pointed CW complexes, the abelian groups $[\Sigma^n X, \Sigma^n Y]$ also become stable under suspensions.

Theorem 7.2.2 Let X and Y be pointed CW complexes. Then the sequence of abelian groups,

$$[\Sigma^n X, \Sigma^n Y] \xrightarrow{\Sigma} [\Sigma^{n+1} X, \Sigma^{n+1} Y] \longrightarrow \cdots$$

are isomorphic under the suspension functor for $n > \dim(X)$ and $n \geq 2$.

New: When X is compact, $[X, QY]_* = [X,Y]^s_*$.

8 Kunneth Theorem for Homology

8.1 The Universal Coefficient Theorem for Homology

Proposition 8.1.1 Let (X, A) be a pair of space. Let T be an abelian group. Then there exists a natural map $h : H_n(X, A) \otimes T \to H_n(X, A; T)$ such that $\text{coker}(h) \cong \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T)$ and a split exact sequence (that is not natural) of the form

$$0 \longrightarrow H_n(X, A) \otimes T \xrightarrow{h} H_n(X, A; T) \longrightarrow \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T) \longrightarrow 0$$

for any $n \in \mathbb{N}$. In particular, split exactness implies that there is an isomorphism

$$H_n(X, A; T) \cong H_n(X, A) \otimes T \oplus \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), T)$$

for any $n \in \mathbb{N}$.

8.2 Kunneth Theorem for Homology

Theorem 8.2.1 (The Eilenberg-Zilbur Theorem)
Let X, Y be spaces. Then $C_\bullet(X) \otimes_{\mathbb{Z}} C_\bullet(Y)$ and $C_\bullet(X \times Y)$ are chain homotopy equivalent, that is natural in X and Y.

Proposition 8.2.2 Let X and Y be CW-complexes. Let R be a principal ideal domain. Then there is a short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X; R) \otimes_R H_j(Y; R) \xrightarrow{\times} H_n(X \times Y; R) \longrightarrow \bigoplus_{i+j=n} \text{Tor}_1^R(H_i(X; R), H_{j-1}(Y; R)) \longrightarrow 0$$

induced by the cross product, that is natural in maps $f : X \to A$ and $g : Y \to B$. Moreover, this sequence splits.

9 Spectral Sequences in Algebraic Topology

9.1 Spectral Sequences in Topology

Theorem 9.1.1 Let X be a space. Let the following be a sequence

$$\emptyset \subset X_0 \subset X_1 \subset \cdots \subset X$$

of subspaces. Let G be an abelian group. Then the following data

- $A_{p,q} = H_{p+q}(X_p; G)$
- $E_{p,q} = H_{p+q}(X_p, X_{p-1}; G)$
- $i : H_{p+q}(X_p; G) = A_{p,q} \to H_{p+q}(X_{p+1}; G) = A_{p+1,q-1}$ (degree $(1, -1)$)
- $j : H_{p+q}(X_p; G) = A_{p,q} \to H_{p+q}(X_p, X_{p-1}; G) = E_{p,q}$ (degree $(0, 0)$)
- $k : H_{p+q}(X_p, X_{p-1}; G) = A_{p,q} \to H_{p+q-1}(X_{p-1}; G) = A_{p-1,q}$ (degree $(-1, 0)$)

defines an exact couple and hence a spectral sequence with E^1 page given by

$$E^1_{p,q} = H_{p+q}(X_p, X_{p-1}; G)$$

where the differential $d : E^1_{p,q} \to E^1_{p-1,q}$ is given by the composition

$$H_{p+q}(X_p, X_{p-1}; G) \xrightarrow{k} H_{p+q-1}(X_{p-1}; G) \xrightarrow{j} H_{p+q-1}(X_{p-1}, X_{p-2}; G)$$

The E_1 page of such a spectral sequence is given by

$$\begin{array}{ccccccc}
\cdots & \longleftarrow & \cdots & \longleftarrow & \cdots & \longleftarrow & \cdots \\
 & & & & & & \\
H_3(X_1, X_0; G) & \longleftarrow & H_4(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) & \longleftarrow & \cdots \\
 & & & & & & \\
H_2(X_1, X_0; G) & \longleftarrow & H_3(X_2, X_1; G) & \longleftarrow & H_4(X_3, X_2; G) & \longleftarrow & \cdots \\
 & & & & & & \\
H_1(X_1, X_0; G) & \longleftarrow & H_2(X_2, X_1; G) & \longleftarrow & H_3(X_3, X_2; G) & \longleftarrow & \cdots
\end{array}$$

Things get interesting when we choose X to be a CW complex and we choose the filtration of X by the skeleton of X. Recall that we have the formula

$$H_{p+q}(X_p, X_{p-1}; G) \cong \begin{cases} C^{\text{CW}}_p(X; G) & \text{if } q = 0 \\ 0 & \text{otherwise} \end{cases}$$

Thus the E^1 page is only left with a chain complex at $q = 0$.

Let us also compute the derived couple of this exact couple or in other words, the E^2 page of the spectral sequence. This is more intuitive then the one thinks about on the definition of the derived couple. The $E^2_{p,q}$ slot is simply the homology of the chain complex at the (p, q)th slot. The direction of the maps of the E^2 page depends not on the choice of spectral sequence at all (In fact, the direction only depends on the page). Now in our case, the homology can be given by a known construct:

$$E^2_{p,q} = \frac{\ker(d : H_{p+q}(X_p, X_{p-1}; G) \to H_{p+q-1}(X_{p-1}, X_{p-2}; G))}{\text{im}(d : H_{p+q+1}(X_{p+1}, X_p; G) \to H_{p+q}(X_p, X_{p-1}; G))} = H^{\text{CW}}_{p+q}(X; G)$$

Since the direction of the maps are now diagonal and when $q \neq 0$ we have $E^2_{p,q} = 0$, all maps in E^2 are 0 and we are left with

$$H^{\text{CW}}_0(X; G) \qquad H^{\text{CW}}_1(X; G) \qquad H^{\text{CW}}_2(X; G) \qquad H^{\text{CW}}_3(X; G) \qquad \cdots$$

Theorem 9.1.2 (Leray-Serre Spectral Sequence) Let $p : E \to B$ be a Serre fibration with fibre F and path connected B. Suppose that the action of $\pi_1(B)$ on $H_*(F; G)$ is trivial. Then there is a first quadrant homological spectral sequence starting with E^2 and weakly converging to $H_\bullet(E; \mathbb{Z})$. Explicitly, there is a convergence

$$E^2_{p,q} = H_p(B, H_1(F)) \Rightarrow H_{p+q}(E; \mathbb{Z})$$

10 Bonus?

10.1 Postnikov Towers

Definition 10.1.1 (Postnikov Towers) Let X be a path connected space. A Postnikov tower is the following commutative diagram

$$\begin{tikzcd}
& X \arrow[d] \arrow[dr] \arrow[drrr] \arrow[drrrr] & & & & & \\
\cdots \arrow[r] & X_n \arrow[r, "p_n"'] & X_{n-1} \arrow[r] & \cdots \arrow[r] & X_2 \arrow[r, "p_2"'] & X_1 \arrow[r, "p_1"'] & *
\end{tikzcd}$$

such that the following are true.

- The maps $X \to X_n$ for each $n \in \mathbb{N}$ induces isomorphisms $\pi_i(X) \cong \pi_i(X_n)$ for $i \leq n$.
- $\pi_i(X_n)$ for $i > n$.
- Each $p_n : X_n \to X_{n-1}$ for $n \in \mathbb{N}$ is a fibration with fiber $K(\pi_n(X), n)$.

Theorem 10.1.2 Suppose that there is an inverse system of spaces

$$\begin{tikzcd}
& \lim_{n\to\infty} X_n \arrow[d] \arrow[dr] \arrow[drrr] \arrow[drrrr] & & & & & \\
\cdots \arrow[r] & X_n \arrow[r, "p_n"'] & X_{n-1} \arrow[r] & \cdots \arrow[r] & X_2 \arrow[r, "p_2"'] & X_1 \arrow[r, "p_1"'] & *
\end{tikzcd}$$

The functor π_i for $i \in \mathbb{N}$ induces a cone in **Grp**. By definition of $\lim_{\leftarrow} \pi_i(X_n)$, there is a unique map

$$\lambda : \pi_i\left(\lim_{\leftarrow} X_n\right) \to \lim_{\leftarrow} \pi_i(X_n)$$

Then the following are true regarding λ.

- λ is surjective
- λ is injective if the maps $\pi_{i+1}(X_n) \to \pi_{i+1}(X_{n-1})$ are surjective for sufficient large n.

Proposition 10.1.3 Let X be a connected CW complex. Then there exists a Postnikov tower for X.

Proposition 10.1.4 Let X be a connected CW complex. Choose a Postnikov tower of X. Then there is a weak homotopy equivalence

$$X \simeq \lim_{\leftarrow} X_n$$

so that X is a CW approximation of $\lim_{\leftarrow} X_n$.